

Experiment No: 4**Experiment Title:** Qualys IP Whitelisting and Connectivity Test**AIM**

To configure firewall rules for approved outbound Qualys HTTPS connectivity and verify them from Kali Linux.

REQUIREMENTS

- Kali Linux
- Internet connection

Step	Command	Purpose
1	<code>sudo apt update</code>	Update package lists
1	<code>sudo apt install ufw dnsutils netcat-openbsd -y</code>	Install UFW, nslookup (dnsutils) and nc
2	<code>ip addr</code>	Check IP address
2	<code>ip route</code>	Check routing
2	<code>ping -c 4 8.8.8.8</code>	Basic connectivity check
2	<code>nslookup google.com</code>	DNS check
3	<code>sudo ufw default deny incoming</code>	Default deny incoming
3	<code>sudo ufw default allow outgoing</code>	Default allow outgoing

3	<code>sudo ufw enable</code>	Enable firewall
4	<code>sudo ufw allow out to <approved-Qualys-IP> port 443 proto tcp</code>	Allow approved Qualys destination on TCP 443
5	<code>sudo ufw status verbose</code>	Verify rules

- UFW
- dnsutils
- netcat-openbsd
- Approved Qualys destinations

PROCEDURE

1. Update Kali and install required tools.
2. Check IP and route using `ip addr` and `ip route`.
3. Set UFW default deny incoming and allow outgoing.
4. Allow approved Qualys destinations on TCP 443.
5. Verify UFW rules.
6. Use `nslookup`, `curl` and `nc` to test connectivity.
7. Enable UFW logging and inspect `journalctl` output.

COMMANDS / CONFIGURATION

Step	Command	Purpose
5	<code>sudo ufw status numbered</code>	Verify rules (numbered)
6	<code>nslookup qualysguard.qg1.apps.qualys.in</code>	Resolve Qualys host
6	<code>curl -I https://qualysguard.qg1.apps.qualys.in</code>	HTTPS test
6	<code>nc -vz qualysguard.qg1.apps.qualys.in 443</code>	TCP 443 test
7	<code>sudo ufw logging medium</code>	Enable UFW logging
7	<code>sudo journalctl -k grep UFW</code>	Inspect UFW log entries

EXPECTED OUTPUT

TCP 443 connectivity to the Qualys server succeeded and UFW logging was verified. The figures below are Expected Output Illustrations generated to represent the expected result; they are not screenshots captured from an actual run.

CONCLUSION

UFW was configured on Kali Linux to allow approved outbound HTTPS (TCP 443) connections to Qualys, and connectivity and logging were verified.